# KOM4521 Multivariable Control Theory

## 2024-2025 Fall

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# Syllabus:

## Week 11-12-13

- State Feedback Controller Design
- Observer Design
- Applications of State Feedback Controllers and Observers

# State Feedback Controller Design

A particular system having dynamics is given as

$$\dot{x}(t) = Ax(t) + Bu(t)$$

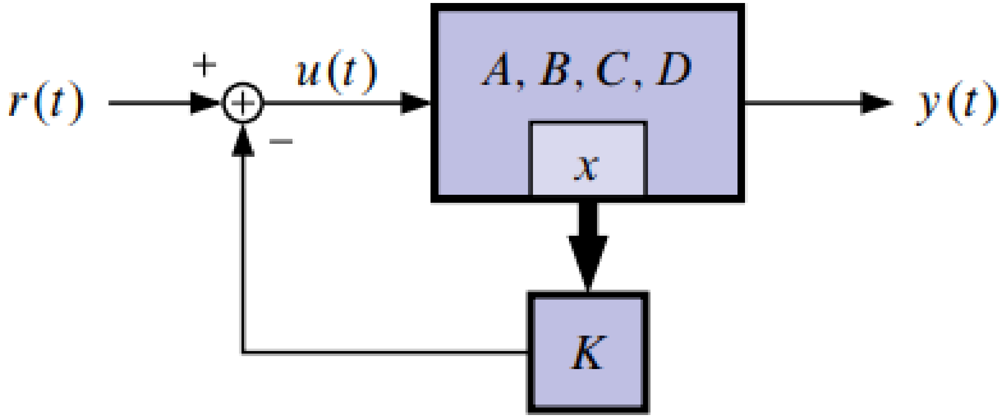
$$y(t) = Cx(t) + Du(t).$$

- Open-loop system poles are given by eigenvalues of $A$.
- Use input $u(t)$ to change the dynamics!

# State Feedback Controller Design

Assume the form of **linear state feedback** with gain vector $\boldsymbol{K}$

$$u(t) = r(t) - Kx(t), \quad K \in \mathbb{R}^{1\times n}.$$



# State Feedback Controller Design

$$\dot{x}(t) = Ax(t) + B(r(t) - Kx(t))$$

$$= (A - BK)x(t) + Br(t)$$

$$y(t) = Cx(t) + Du(t).$$

**OBJECTIVE:** Design $K$ so that $A_{CL} = A - BK$ has some nice properties. For example,

- Perhaps $A$ is unstable. Design $A_{CL}$ to be stable.
- Or, pole placement.

# State Feedback Controller Design

Example: Consider the (unstable) system

$$\dot{x}(t) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t).$$

$$\det(sI - A) = (s-1)(s-2) - 1 = s^2 - 3s + 1.$$

- Let

$$u(t) = -\begin{bmatrix} k_1 & k_2 \end{bmatrix} x(t) = -Kx(t)$$

$$\begin{aligned} A_{CL} = A - BK &= \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} k_1 & k_2 \end{bmatrix} \\ &= \begin{bmatrix} 1-k_1 & 1-k_2 \\ 1 & 2 \end{bmatrix}. \end{aligned}$$

# State Feedback Controller Design

- So, $\det(sI - A_{CL}) = s^2 + (k_1 - 3)s + (1 - 2k_1 + k_2)$.
- By choosing $k_1$ and $k_2$, we can put $\mathrm{eig}(A_{CL})$ *anywhere* in the complex plane (in complex-conjugate pairs, that is!)
- For example, how can we place poles at $-5$, $-6$?
- Compare desired closed-loop characteristic equation

$$(s + 5)(s + 6) = s^2 + 11s + 30$$

with

$$\det(sI - A_{CL}) = s^2 + (k_1 - 3)s + (1 - 2k_1 + k_2).$$

- So,

$$k_1 - 3 = 11, \quad \text{or, } k_1 = 14$$

$$1 - 2k_1 + k_2 = 30, \quad \text{or, } k_2 = 57.$$

- $K = [14\ 57]$.

# State Feedback Controller Design

Example: Consider the system

$$\dot{x}(t) = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t) \qquad \text{and} \qquad u(t) = -Kx(t).$$

$$A_{CL} = A - BK = \begin{bmatrix} 1-k_1 & 1-k_2 \\ 0 & 2 \end{bmatrix}$$

$$\det(sI - A_{CL}) = (s-1+k_1)(s-2).$$

- Feedback of the state cannot move the pole at $s = 2$. System *cannot* be stabilized via state feedback. (Note that the system is already in Kalman form, and the uncontrollable mode has eigenvalue 2).

# Bass–Gura pole placement

If $\{A, B\}$ is controllable, then we can arbitrarily assign the eigenvalues of $A_{CL}$ using state feedback. More precisely, given any polynomial $s^n + \alpha_1 s^{n-1} + \cdots + \alpha_n$ there exists a (unique for SISO) $K \in \mathbb{R}^{m \times n}$ ($m$ = number of system inputs = 1 for SISO such that

$$\det(sI - A + BK) = s^n + \alpha_1 s^{n-1} + \cdots + \alpha_n .$$

# Bass–Gura pole placement

If $\{A, B\}$ is controllable, we can find a state feedback for which the closed-loop realization is stable.

- Transform $\{A, B, C, D\}$ into controller canonical form. (Then solve problem in this form, then convert back to original form).

# Bass–Gura pole placement

- That is, find $T$ such that

$$T^{-1}AT = A_c = \begin{bmatrix} -a_1 & -a_2 & \cdots & -a_n \\ 1 & & & 0 \\ & \ddots & & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix} \quad \text{and} \quad T^{-1}B = B_c = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

where $\det(sI - A) = s^n + a_1 s^{n-1} + \cdots + a_n$.

# Bass–Gura pole placement

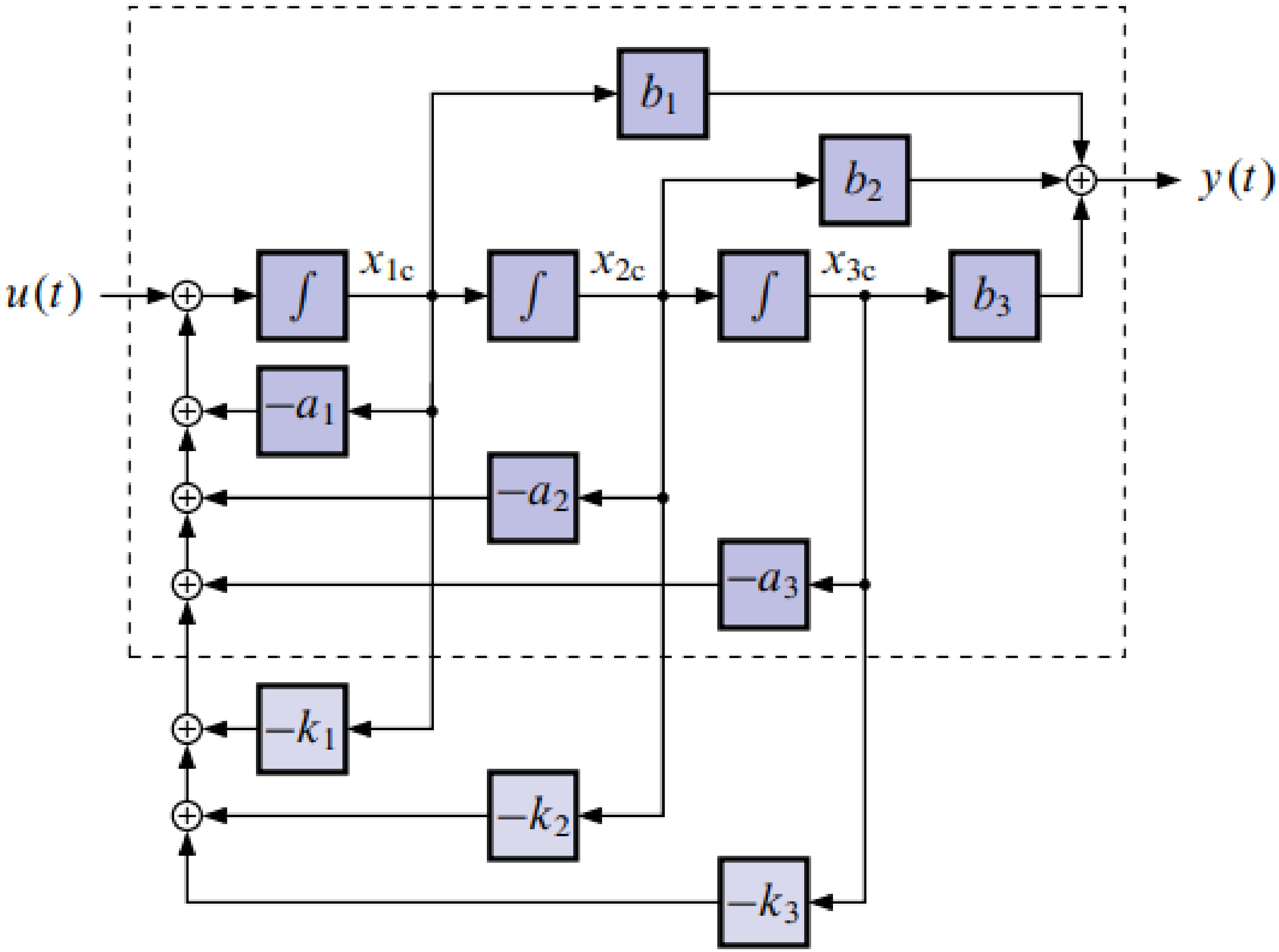
- Let's apply state-feedback $K_c$ to the controller realization.
- Note, $K_c = \begin{bmatrix} k_1 & \cdots & k_n \end{bmatrix}$, so

$$B_c K_c = \begin{bmatrix} k_1 & k_2 & \cdots & k_n \\ 0 & & & 0 \\ \vdots & & & \vdots \\ 0 & & & 0 \end{bmatrix}.$$

- Useful because characteristic equation obvious.

$$A_{CL} = A_c - B_c K_c = \begin{bmatrix} -(a_1 + k_1) & -(a_2 + k_2) & \cdots & -(a_n + k_n) \\ 1 & & & 0 \\ & \ddots & & \vdots \\ 0 & \cdots & 1 & 0 \end{bmatrix},$$

# Bass–Gura pole placement



# Bass–Gura pole placement

- Thus, after state feedback with $K_c$ the characteristic equation is

$$\det(sI - A_c + B_c K_c) = s^n + (a_1 + k_1)s^{n-1} + \cdots + (a_n + k_n).$$

- The desired characteristic equation is

$$\chi_d(s) = s^n + \alpha_1 s^{n-1} + \cdots + \alpha_n.$$

- Equating coefficients of powers-of-$s$, we set $k_1 = \alpha_1 - a_1, \ldots, k_n = \alpha_n - a_n$ and get the desired characteristic polynomial.

# Bass–Gura pole placement

- Now that we have the solution in the controller canonical form, we transform back to the original realization

$$
\begin{aligned}
\chi_d(s) &= \det(sI - A_c + B_c K_c) \\
&= \det(T)\det(sI - A_c + B_c K_c)\det(T^{-1}) \\
&= \det(sI - TA_cT^{-1} + TB_cK_cT^{-1}) \\
&= \det(sI - A + BK_cT^{-1}).
\end{aligned}
$$

$$\chi(s) = \det(sI - A + BK) \qquad \text{so}$$

$$K = K_cT^{-1}.$$

So, if we use state feedback

$$
\begin{aligned}
K &= K_cT^{-1} \\
&= \begin{bmatrix} (\alpha_1 - a_1) & \cdots & (\alpha_n - a_n) \end{bmatrix} T^{-1}
\end{aligned}
$$

we will have the desired characteristic polynomial.

# Bass–Gura pole placement

- One remaining question: What is $T$? We know $T = \mathcal{C}\mathcal{C}_c^{-1}$ and

$$\mathcal{C}_c = \begin{bmatrix} 1 & -a_1 & a_1^2 - a_2 & \cdots \\ 0 & 1 & -a_1 & 0 \\ \vdots & & 1 & \vdots \\ 0 & \cdots & & 1 \end{bmatrix}$$

$$\mathcal{C}_c^{-1} = \underbrace{\begin{bmatrix} 1 & a_1 & \cdots & a_{n-1} \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & a_1 \\ 0 & & & 1 \end{bmatrix}}_{\text{upper } \Delta \text{ Toeplitz}}$$

# Bass–Gura pole placement

$$T^{-1} = \begin{bmatrix} 1 & a_1 & \cdots & a_{n-1} \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & a_1 \\ 0 & & & 1 \end{bmatrix}^{-1} \mathcal{C}^{-1}$$

- So,

$$K = \begin{bmatrix} (\alpha_1 - a_1) & \cdots & (\alpha_n - a_n) \end{bmatrix} \begin{bmatrix} 1 & a_1 & \cdots & a_{n-1} \\ 0 & 1 & & a_{n-2} \\ \vdots & & \ddots & \vdots \\ 0 & & & 1 \end{bmatrix}^{-1} \mathcal{C}^{-1}.$$

- This is called the *Bass–Gura formula* for $K$.

# Ackermann's formula

Consider (a system already in controller canonical form) $\chi(s) = s^n + a_1 s^{n-1} + \cdots + a_n s^0$

$$\chi(A_c) = A_c^n + a_1 A_c^{n-1} + \cdots + a_n I = 0$$

by Cayley-Hamilton. Also,

$$\chi_d(s) = s^n + \alpha_1 s^{n-1} + \cdots + \alpha_n s^0$$

$$\chi_d(A_c) = A_c^n + \alpha_1 A_c^{n-1} + \cdots + \alpha_n I \neq 0$$

$$\chi_d(A_c) - 0 = \chi_d(A_c) - \chi(A_c) = (\alpha_1 - a_1) A_c^{n-1} + \cdots + (\alpha_n - a_n) I.$$

# Ackermann's formula

A sneaky fact for the controller form is

$$\begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} A_c^k = \begin{bmatrix} 0 & \cdots & \underbrace{1}_{n-k} & \cdots & 0 \end{bmatrix}$$

so that

$$\begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} \chi_d(A_c) = \begin{bmatrix} (\alpha_1 - a_1) & (\alpha_2 - a_2) & \cdots & (\alpha_n - a_n) \end{bmatrix} = K_c$$

$$K_c = \begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} \chi_d(A_c).$$

# Ackermann's formula

To convert to the original form,

$$
\begin{aligned}
K &= K_c T^{-1} \\
&= \begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} \chi_d(T^{-1}AT)T^{-1} \\
&= \begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} T^{-1}\chi_d(A) \\
&= \begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} \mathcal{C}_c\mathcal{C}^{-1}\chi_d(A) \\
&= \begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} \mathcal{C}^{-1}\chi_d(A).
\end{aligned}
$$

# Ackermann's formula

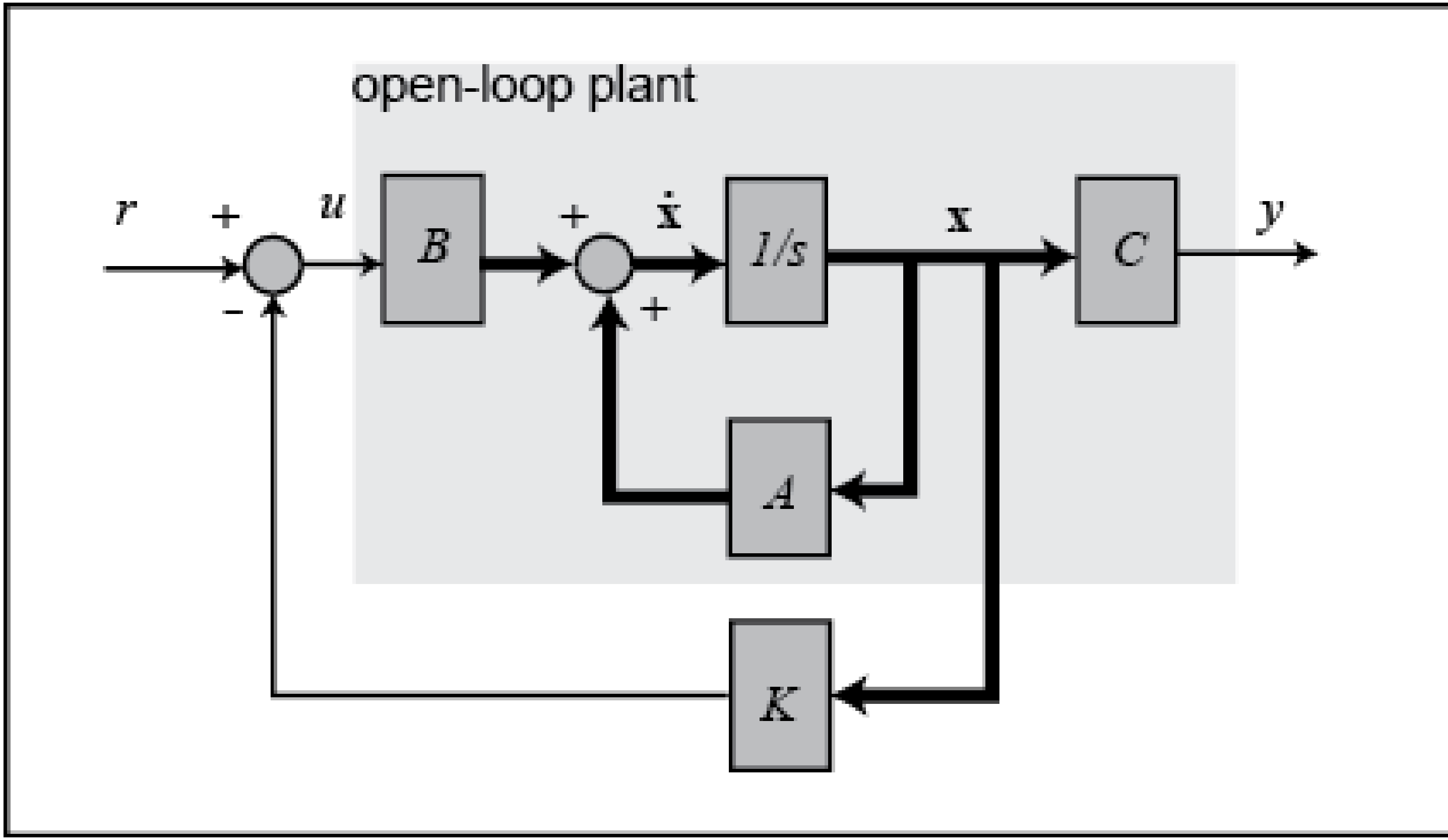
- Revisit previous example. $\chi_d(s) = s^2 + 11s + 30.$

$$\mathcal{C} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned} K &= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} + 11 \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} + 30 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} \\ &= \begin{bmatrix} 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} + \begin{bmatrix} 41 & 11 \\ 11 & 52 \end{bmatrix} \right\} \\ &= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 43 & 14 \\ 14 & 57 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 57 \end{bmatrix}. \end{aligned}$$

✓ same as before.

# State-feedback Controller Design



# State-feedback Controller Design

- The eigenvalues associated with uncontrollable modes are fixed (don't change) under state feedback, but those associated with controllable modes can be arbitrarily assigned.
- State feedback does not change zeros of a realization.

# State-feedback Controller Design

- Drastic changes in characteristic polynomial requires large gains $K$ (high control effort).
- State feedback can result in unobservable modes (pole-zero cancellations).

# State-feedback Controller Design

## Example:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad K = \begin{bmatrix} 14 & 57 \end{bmatrix}.$$

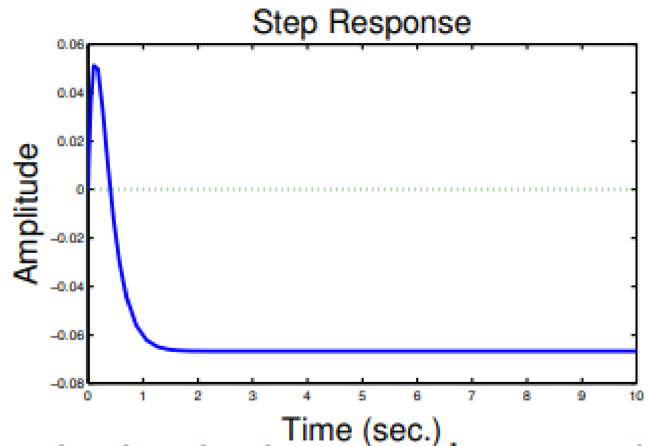
- Let $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$. Then $\dfrac{Y(s)}{R(s)} = C(sI - A + BK)^{-1}B.$

$$\frac{Y(s)}{R(s)} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s+13 & 56 \\ -1 & s-2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{\begin{bmatrix} s-2 & -56 \\ 1 & s+13 \end{bmatrix}}{s^2 + 11s - 26 + 56} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{s-2}{s^2 + 11s + 30}.$$

- Final value theorem for step input, $y(t) \to \dfrac{-2}{30} \neq 1$ !

# State-feedback Controller Design




**OBSERVATION:** A constant output $y_{ss}$ requires constant state $x_{ss}$ and constant input $u_{ss}$. We can change the tracking problem to a regulation problem around $u(t) = u_{ss}$ and $x(t) = x_{ss}$.

$$(u(t) - u_{ss}) = -K(x(t) - x_{ss}).$$

- $u_{ss}$ and $x_{ss}$ related to $r_{ss}$. Let

$$u_{ss} = \underbrace{N_u}_{1\times 1} r_{ss}$$

$$x_{ss} = \underbrace{N_x}_{n\times 1} r_{ss}.$$

# State-feedback Controller Design

- How to find $N_u$ and $N_x$? Use equations of motion.

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t).$$

- At steady state,

$$\dot{x}_{ss}(t) = 0 = Ax_{ss} + Bu_{ss} = [AN_x + BN_u]r_{ss}$$

$$y_{ss}(t) = r_{ss} = Cx_{ss} + Du_{ss} = [CN_x + DN_u]r_{ss}.$$

- Two equations and two unknowns.

$$\left[\begin{array}{c:c} A & B \\ \hdashline C & D \end{array}\right]\left[\begin{array}{c} N_x \\ \hdashline N_u \end{array}\right] = \left[\begin{array}{c} 0 \\ \hdashline I \end{array}\right].$$

# State-feedback Controller Design

- In steady-state we had

$$(u(t) - u_{ss}) = -K(x(t) - x_{ss})$$

which is achieved by the control signal

$$\begin{aligned} u(t) &= N_u r(t) - K(x(t) - N_x r(t)) \\ &= -Kx(t) + (N_u + KN_x) r(t) \\ &= -Kx(t) + \bar{N} r(t). \end{aligned}$$

- $\bar{N}$ computed without knowing $r(t)$. It works for any $r(t)$.
- In our example we can find that

$$\begin{bmatrix} N_x \\ \hdashline N_u \end{bmatrix} = \begin{bmatrix} 1 \\ -1/2 \\ \hdashline -1/2 \end{bmatrix}.$$

- $N_u + KN_x = -15.$

# State-feedback Controller Design

- New equations:

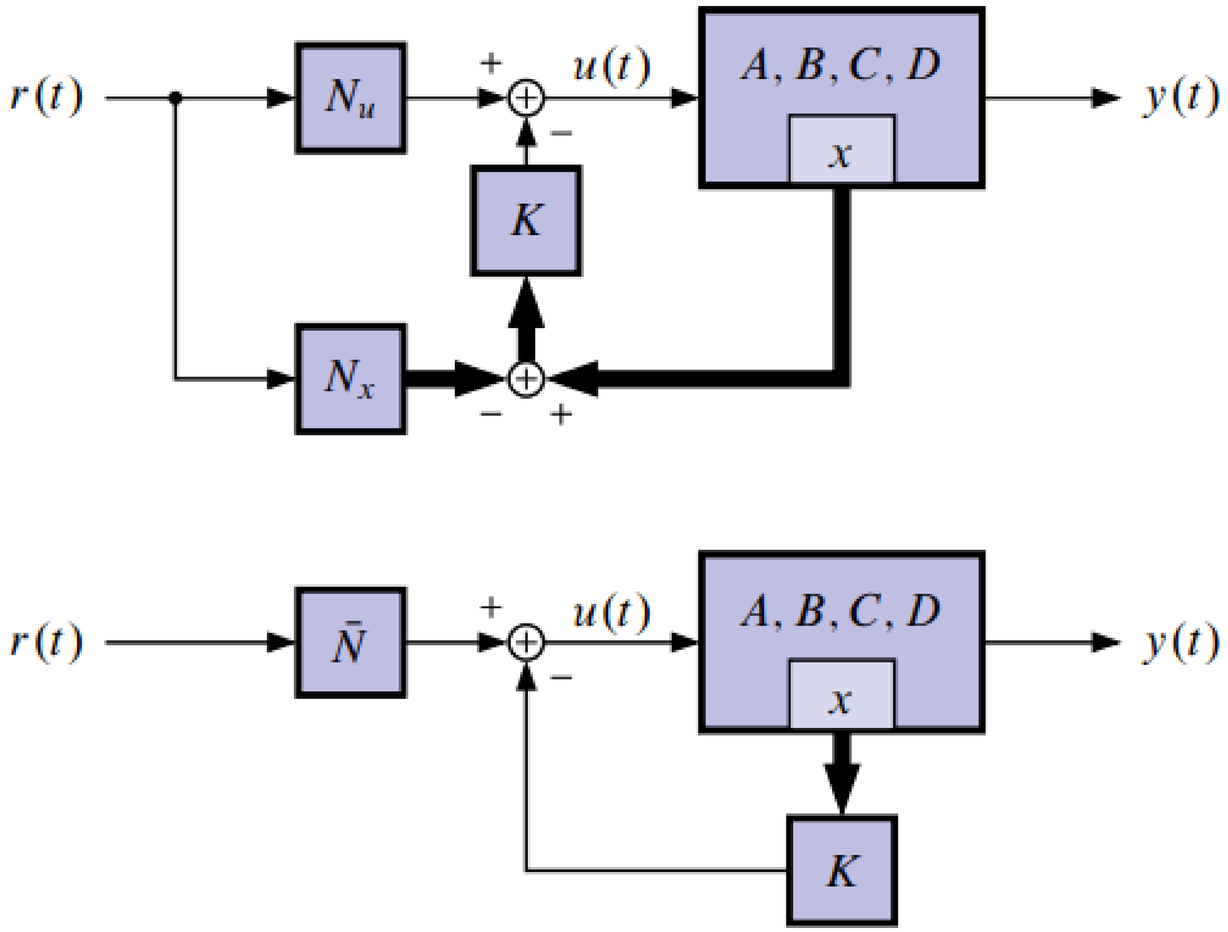
$$\begin{aligned}\dot{x}(t) &= Ax(t) + B(N_u + KN_x)r(t) - BKx(t)\\ &= (A - BK)x(t) + B\bar{N}r(t)\\ y(t) &= Cx(t).\end{aligned}$$

- Therefore,

$$\left(\frac{Y(s)}{R(s)}\right)_{\text{new}} = \left(\frac{Y(s)}{R(s)}\right)_{\text{old}} \times \bar{N} = \frac{-15s + 30}{s^2 + 11s + 30}$$

which has zero steady-state error to a unit-step.

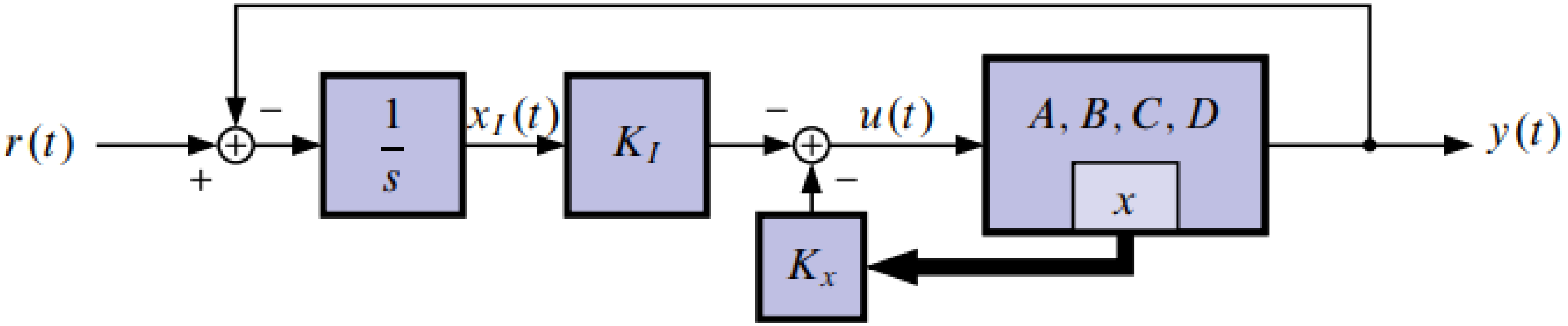
# State-feedback Controller Design



# Integral control for continuous-time systems

- In many practical designs, integral control is needed to counteract disturbances, plant variations, or other noises in the system.
- Up until now, we have not seen a design that has integral action. In fact state-space designs will NOT produce integral action unless we make special steps to include it!

# Integral control for continuous-time systems




$$
\begin{aligned}
\dot{x}_I(t) &= r(t) - y(t) \\
&= r(t) - (Cx(t) + Du(t)).
\end{aligned}
$$

# Integral control for continuous-time systems

Note that we can include the integral state into our normal state-space form by augmenting the system dynamics.

$$\begin{bmatrix} \dot{x}_I(t) \\ \hdashline \dot{x}(t) \end{bmatrix} = \left[\begin{array}{c:c} 0 & -C \\ \hdashline 0 & A \end{array}\right] \begin{bmatrix} x_I(t) \\ \hdashline x(t) \end{bmatrix} + \begin{bmatrix} -D \\ \hdashline B \end{bmatrix} u(t) + \begin{bmatrix} I \\ \hdashline 0 \end{bmatrix} r(t)$$

$$y(t) = Cx(t) + Du(t).$$

# Integral control for continuous-time systems

Note that the matrix that now fills the state-space placeholder usually called "$A$" has an open-loop eigenvalue at the origin. This corresponds to increasing the system type, and integrates out steady-state error.

■ The control law is,

$$u(t) = -\left[\begin{array}{c:c} K_I & K_x \end{array}\right]\left[\begin{array}{c} x_I(t) \\ \hdashline x(t) \end{array}\right].$$

# Integral control for continuous-time systems

This is state feedback on the augmented state vector. We can find the feedback gain vector $K$ by using Ackerman's formula (or similar) by replacing "$A$" in Ackerman with the augmented "$A$" matrix above, and by replacing "$B$" in Ackerman by the matrix multiplying $u(t)$ above.

# Integral control for continuous-time systems

■ Note that the augmented system has $n + n_I$ open-loop poles, so we will have to choose $n + n_I$ desired closed-loop poles, and split the resulting $K$ into the parts $K_I$ and $K_x$ .

When we substitute the control law for $u(t)$ into the open-loop augmented state-space dynamics, we get:

# Integral control for continuous-time systems

■ This is a state-space system with the augmented state vector, and input $r(t)$.

$$\begin{bmatrix} \dot{x}_I(t) \\ \hdashline \dot{x}(t) \end{bmatrix} = \left( \overbrace{\underbrace{\begin{bmatrix} 0 & -C \\ 0 & A \end{bmatrix}}_{A \text{ in Acker/place}} - \underbrace{\begin{bmatrix} -D \\ \hdashline B \end{bmatrix}}_{B \text{ in Acker/place}} \begin{bmatrix} K_I \,\vdots\, K_x \end{bmatrix}}^{A_{CL}} \right) \begin{bmatrix} x_I(t) \\ \hdashline x(t) \end{bmatrix} + \begin{bmatrix} I \\ \hdashline 0 \end{bmatrix} r(t)$$

# Observer design

■ State feedback is impractical since we don't know the state! But, what if we can estimate the state?

- ➢ If $x(t)$ is the true state, $\hat{x}(t)$ is called the state estimate.
- ➢ We want $x(t) = \hat{x}(t)$, or at least $x(t) \rightarrow \hat{x}(t)$. How do we build this?

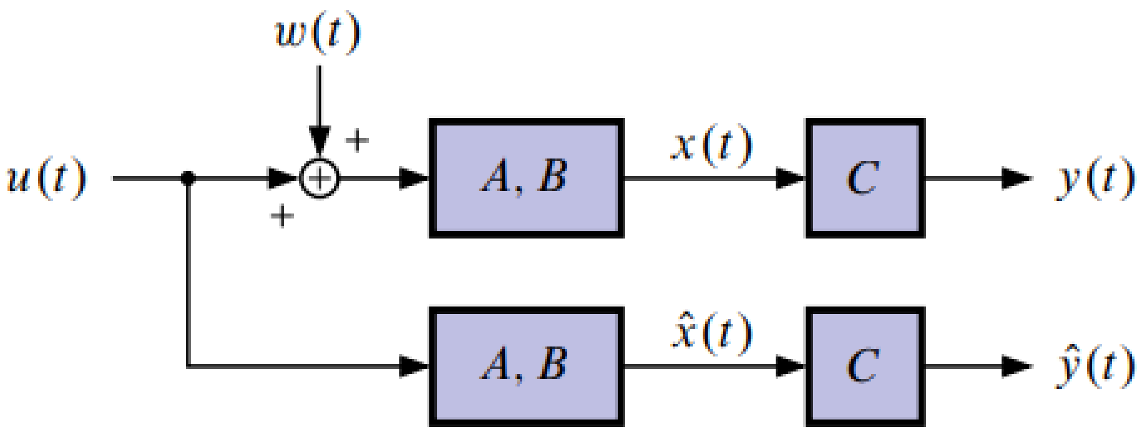
# Observer design

- To start, use our knowledge of the plant

$$\dot{x}(t) = Ax(t) + Bu(t).$$

- Let our state estimate be

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t).$$



# Observer design

- This is called an "open-loop estimator."
- Let's analyze our open-loop estimator by examining the state-estimate error

$$\tilde{x}(t) = x(t) - \hat{x}(t).$$

- We want $\tilde{x}(t) = 0$ . For our estimator

$$\begin{aligned}
\dot{\tilde{x}}(t) &= \dot{x}(t) - \dot{\hat{x}}(t) \\
&= Ax(t) + Bu(t) - A\hat{x}(t) - Bu(t) \\
&= A\tilde{x}(t).
\end{aligned}$$

## Observer design

- So,

$$\tilde{x}(t) = e^{At}\tilde{x}(0).$$

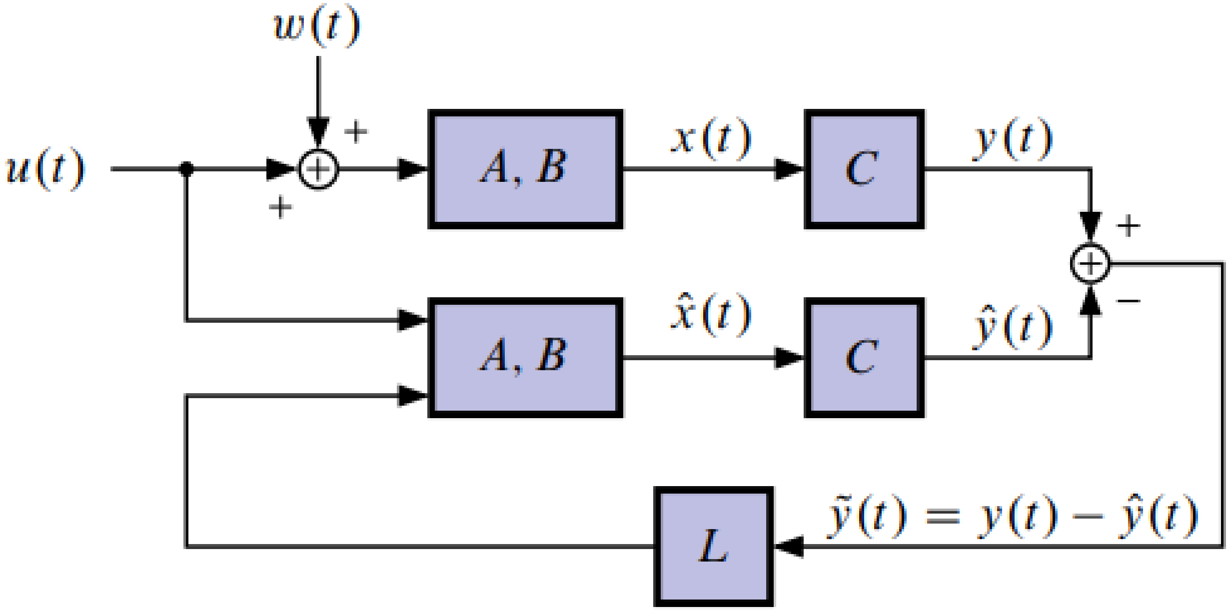
- Hence, $x(t) \to \hat{x}(t)$ if $A$ is stable! (This is not too impressive though since $x(t) \to \hat{x}(t)$ because both $\hat{x}(t)$ and $x(t)$ go to zero).

# Observer design

We need to improve our estimator:

- Speed up convergence.
- Reduce sensitivity to model uncertainties.
- Counteract disturbances.
- Have convergence even when A is unstable.

# Observer design




- This is called a "closed-loop estimator" (assuming $\hat{y} = C\hat{x}$). If $L = 0$ we have an open-loop estimator.

# Observer design

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L\left(y(t) - \hat{y}(t)\right).$$

$$\begin{aligned}
\dot{\tilde{x}}(t) &= \dot{x}(t) - \dot{\hat{x}}(t) \\
&= Ax(t) + Bu(t) - A\hat{x}(t) - Bu(t) - L\left(y(t) - C\hat{x}(t)\right) \\
&= A\tilde{x}(t) - L\left(Cx(t) - C\hat{x}(t)\right) \\
&= (A - LC)\,\tilde{x}(t);
\end{aligned}$$

$$\tilde{x}(t) = e^{(A-LC)t}\tilde{x}(0),$$

$x(t) \rightarrow \hat{x}(t)$ if $A - LC$ is stable, for any value of $\hat{x}(0)$ and any $u(t)$,whether or not $A$ is stable.

# Observer design

In fact, we can look at the dynamics of the state estimate error to quantitatively evaluate how $\hat{x}(t) \rightarrow x(t)$.

$$\dot{\tilde{x}}(t) = (A - LC)\,\tilde{x}(t)$$

has dynamics related to the roots of the characteristic equation

$$\chi_{ob}(s) = \det(sI - A + LC) = 0.$$

# Observer design

So, for our estimator, we specify the convergence rate of $\hat{x}(t) \rightarrow x(t)$ by choosing desired pole locations: Choose $L$ such that

$$\chi_{ob,des}(s) = \det(sI - A + LC).$$

- This is called the "observer gain problem".

# Bass–Gura inspired method

- We want a specific characteristic equation for $A - LC$.
- Suppose $\{C, A\}$ is observable. Put $\{A, B, C, D\}$ in observer canonical form.
- That is, find $T$ such that

$$T^{-1}AT = A_o = \begin{bmatrix} -a_1 & 1 & & 0 \\ -a_2 & & \ddots & \vdots \\ \vdots & & & 1 \\ -a_n & 0 & \cdots & 0 \end{bmatrix} \quad \text{and} \quad CT = C_o = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix}$$

where $\det(sI - A) = s^n + a_1 s^{n-1} + \cdots + a_n$.

# Bass–Gura inspired method

- Apply feedback $\mathrm{L}_0$ to the observer realization to end up with $\mathrm{A}_0 - \mathrm{L}_0\mathrm{C}_0$.
- Note,

$$L_o = \begin{bmatrix} l_1 & \cdots & l_n \end{bmatrix}^T, \text{ so}$$

$$L_o C_o = \begin{bmatrix} l_1 & 0 & \cdots & 0 \\ l_2 & & & \\ \vdots & & & \\ l_n & 0 & \cdots & 0 \end{bmatrix}.$$

# Bass–Gura inspired method

$$A_o - L_o C_o = \begin{bmatrix} -(a_1 + l_1) & 1 & & 0 \\ -(a_2 + l_2) & & \ddots & \vdots \\ \vdots & & & 1 \\ -(a_n + l_n) & 0 & \cdots & 0 \end{bmatrix},$$

Still in observer form!

- After feedback with $L_0$ the characteristic equation is

$$\det(sI - A_o + L_o C_o) = s^n + (a_1 + l_1)s^{n-1} + \cdots + (a_n + l_n).$$

# Bass–Gura inspired method

- If we set $l_1 = \alpha_1 - a_1, \ldots, l_n = \alpha_n - a_n$ then we get the desired characteristic polynomial.
- Now, we transform back to the original realization

$$\det(sI - A_o + L_oC_o) = \det(sI - TA_oT^{-1} + TL_oC_oT^{-1})$$

$$= \det(sI - A + TL_oC).$$

# Bass–Gura inspired method

- So, if we use feedback

$$L = TL_o$$
$$= T\left[\ (\alpha_1 - a_1)\ \cdots\ (\alpha_n - a_n)\ \right]^T$$

- we will have the desired characteristic polynomial.
- What is $T$ ?

# Bass–Gura inspired method

- What is $T$ ?

$$T = \mathcal{O}^{-1}\mathcal{O}_o$$

$$\mathcal{O}_o = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ a_1 & 1 & & \vdots \\ \vdots & & \ddots & 0 \\ a_{n-1} & \cdots & a_1 & 1 \end{bmatrix}^{-1}$$

$$L = \mathcal{O}^{-1} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ a_1 & 1 & & \vdots \\ \vdots & & \ddots & 0 \\ a_{n-1} & \cdots & a_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} (\alpha_1 - a_1) \\ (\alpha_2 - a_2) \\ \vdots \\ (\alpha_n - a_n) \end{bmatrix}.$$

- This is the Bass–Gura formula for $L$.

# Bass–Gura inspired method: Duality

Notice that if you $A \leftarrow A^T, B \leftarrow C^T, L \leftarrow K^T$ substitute in the Bass–Gura controller design procedure, you will get exactly this equation. Why?

- Notice that we wish to design certain pole locations: $\text{eig}(A - LC) = \text{eig}(A^T - C^T L^T)$ which has the same form as the controller design problem $\text{eig}(A - BK)$ if we make the above substitutions.

# The Ackermann-inspired method

The observer-gain problem is “dual” to the controller-gain problem. Replace a controller-design procedure’s inputs

$$A \leftarrow A^T, B \leftarrow C^T, K \leftarrow L^T$$

Design the controller. Then, $L$ will be the observer gains.

$$K = \begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} \mathcal{C}(A,B)^{-1} \chi_d(A)$$

$$L = \left[ \begin{bmatrix} 0 & \cdots & 1 \end{bmatrix} \mathcal{C}(A,B)^{-1} \chi_d(A) \right]^T_{\substack{A \leftarrow A^T \\ B \leftarrow C^T}}$$

# The Ackermann-inspired method

$$= \chi_d^T(A^T) \begin{bmatrix} C^T & A^T C^T & \cdots & {A^{n-1}}^T C^T \end{bmatrix}^{-T} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$= \chi_d(A) \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$= \chi_d(A) \mathcal{O}(C, A)^{-1} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

# Separation principle

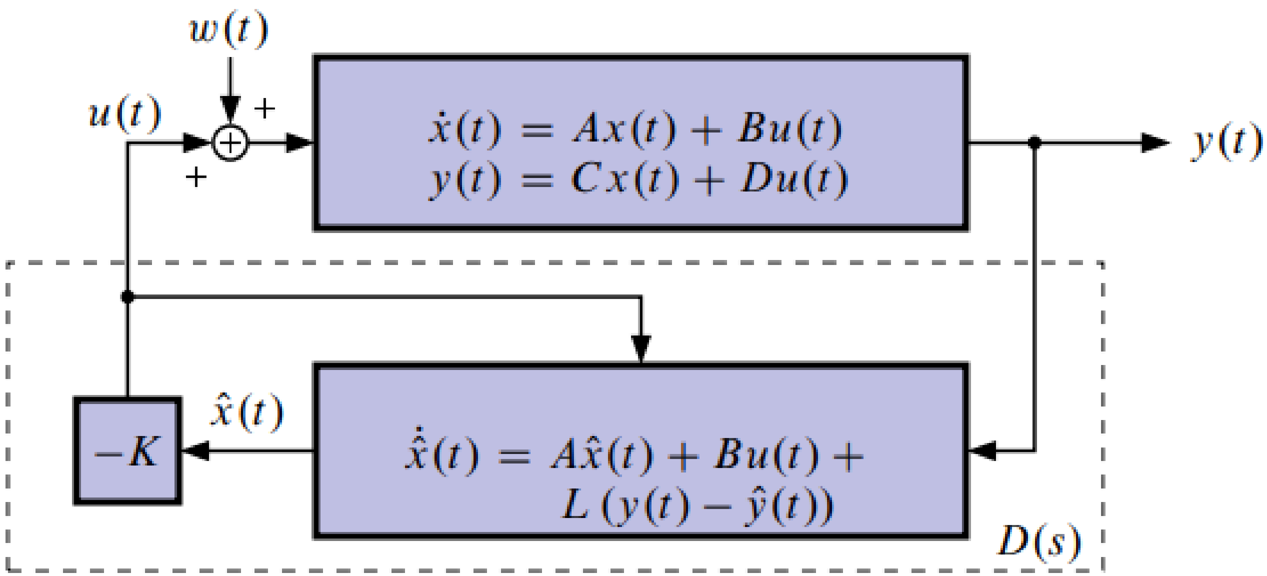
Now that we have a structure to estimate the state $x(t)$, let's feed back $\hat{x}(t)$ to control the plant. That is,

$$u(t) = r(t) - K\hat{x}(t),$$

where $K$ was designed assuming that $u(t) = r(t) - Kx(t)$. Is this going to work? How risky is it to interconnect two well-behaved, stable systems?

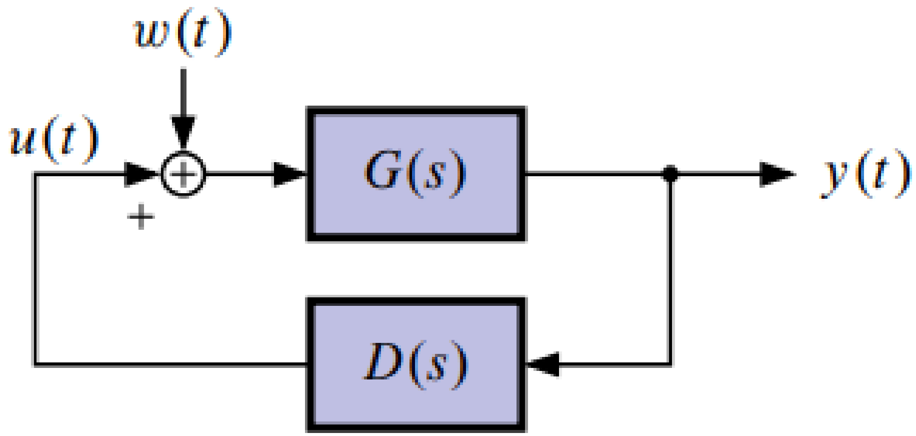
(Assume $r(t) = 0$ for now).

# Separation principle




What is inside the dotted line is equivalent to our classical design compensator →

# Separation principle




where $G(s)$ is described by

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t),$$

and $D(s)$ is described by

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L\,(y(t) - \hat{y}(t))$$

$$u(t) = -K\hat{x}(t),$$

# Separation principle

$$\dot{x}(t) = Ax(t) - BK\hat{x}(t)$$

$$\begin{aligned}\dot{\hat{x}}(t) &= (A - BK)\hat{x}(t) + L(Cx(t) + Du(t) - C\hat{x}(t) - Du(t)) \\ &= (A - BK - LC)\hat{x}(t) + LCx(t).\end{aligned}$$

$$\left[\begin{array}{c} \dot{x}(t) \\ \hdashline \dot{\hat{x}}(t) \end{array}\right] = \left[\begin{array}{c:c} A & -BK \\ \hdashline LC & A - BK - LC \end{array}\right] \left[\begin{array}{c} x(t) \\ \hdashline \hat{x}(t) \end{array}\right],$$

in terms of $\tilde{x}$ (t) →

# Separation principle

$$\dot{\tilde{x}}(t) = \dot{x}(t) - \dot{\hat{x}}(t)$$

$$= Ax(t) - BK\hat{x}(t) - LCx(t) - (A - BK - LC)\,\hat{x}(t)$$

$$= (A - LC)\,\tilde{x}(t)$$

$$\dot{x}(t) = Ax(t) - BK\,(x(t) - \tilde{x}(t))\,,$$

$$\begin{bmatrix} \dot{x}(t) \\ \hline \dot{\tilde{x}}(t) \end{bmatrix} = \left[\begin{array}{c|c} A - BK & BK \\ \hline 0 & A - LC \end{array}\right] \begin{bmatrix} x(t) \\ \hline \tilde{x}(t) \end{bmatrix}.$$

# Separation principle

Note, we have simply changed coordinates:

$$\begin{bmatrix} x(t) \\ \hline \hat{x}(t) \end{bmatrix} = \underbrace{\left[\begin{array}{c:c} I & 0 \\ \hdashline I & -I \end{array}\right]}_{T} \begin{bmatrix} x(t) \\ \hline \tilde{x}(t) \end{bmatrix}.$$

with this change of coordinates, we may easily find the closed-loop poles of the combined state-feedback/estimator system: The eigenvalues of a block-upper-triangular $2x2$ matrix are the collection of the eigenvalues of the two diagonal blocks.

# Separation principle

- Therefore, the $2n$ poles are the $n$ eigenvalues of $A - BK$ combined with the n eigenvalues of $A - LC$.
- But, we designed the $n$ eigenvalues of $A - BK$ to give good (stable) state-feedback performance, and we designed the n eigenvalues of $A - LC$ to give good (stable) estimator performance. Therefore, the closed-loop system is also stable.

# Separation principle

- This is such an astounding conclusion that it has been given a special name: The "Separation Principle".
- It implies that we can design a compensator in two steps:
  1. Design state-feedback assuming the state $x(t)$ is available.
- 2. Design the estimator to estimate the state as $\hat{x}(t)$ and, that $\mathrm{u(t)} = -K\hat{x}(t)$ works!